

1(a)

one mark max for each feature

3

feature common to bacterial and plant cells	bacterial cell	plant cell
cell surface membrane	contain hopanoids	contains cholesterol
ribosome	70S / smaller / 18-20nm I no 80S ribosomes	80S / larger / 22-30nm ; A and 70S
DNA max 1 mark must be across the dotted rows	circular / closed loop I ref. to plasmids	linear ; AW
	naked A no histones	(complexed) with, histones / (basic) proteins ;
	free in, cytoplasm / cytosol A not enclosed / AW, by nuclear envelope A found in nucleoid region	enclosed by / AW, nuclear envelope ; A in a nucleus
cell wall	(mainly composed of) peptidoglycan / murein	(mainly composed of) cellulose ;

Question	Answer	Marks
1(b)	any three from: A particles <i>for substances</i> I <i>ref. to size</i> 1 for transport of ions <u>and</u>, hydrophilic / polar, molecules / substances AW or for transport of substances that cannot pass (directly) across the (phospholipid) bilayer / hydrophobic core ; AW R if implied that all substances cannot cross the bilayer I substances, bypass / do not need, to cross the bilayer 2 carrier proteins for active transport and facilitated diffusion <u>and</u> channel proteins for facilitated diffusion ; 3 further detail carrier protein ; I water-filled / hydrophilic interior e.g. have specific binding sites are specific in, molecule / ion / named example, transported AW carry out conformational change (following binding) / AW use, ATP / energy active transport) moves substances against the concentration gradient / AW (active transport) moves substances down the concentration gradient / AW (facilitated diffusion) <i>accept once in mp3 or 4</i> I moves substances by diffusion 4 further detail channel protein ; e.g. hydrophilic / water-filled, pore / channel A hydrophilic lining some, ion channels / proteins channels for ions, are selective (for size and charge) (most) channel proteins are not selective aquaporins for passage of (larger quantities of) water some, are gated / voltage-gated moves substances down the concentration gradient / AW	3

Question	Answer	Marks
1(c)	1 ref. to light absorption any one from: <i>chloroplasts at periphery because</i> pushed by vacuole ; vacuole is, turgid / filled with cell sap ; (large permanent) vacuole is in the centre (of the cell) ;	1
1(d)	metaphase label to metaphase cell ; prophase label to one of the prophase cells ;	2

Question	Answer	Marks
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